



INDIA.RUNS

Build what next India runs on

Track 1

AI Systems & Workflow Innovation Challenge

Redrob Context & Ideathon Scope

Redrob AI is building an AI-native ecosystem spanning hiring, productivity, communication, discovery, workflows, recommendations, automation, and intelligent user experiences.

Participants are encouraged to think beyond standalone applications and explore how their ideas can:

Extend existing
Redrob capabilities

Improve experiences
for Redrob users

Introduce new
AI-native
workflows

Strengthen ecosystem
engagement and network
effects

Important

Participants are not expected to build a completely new platform from scratch.

Strong submissions will demonstrate:

How the idea
leverages existing
Redrob
capabilities

What new
capability,
workflow, or
value layer is
introduced

How the idea
strengthens the
overall Redrob
ecosystem

Why Redrob is
uniquely positioned to
enable this experience

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Track 01

AI Systems & Workflow Innovation Challenge

Team Name : Tech Adi

Problem Statement : Recruiters spend 60%+ of time manually screening candidates.
Keyword-based ATS misses qualified candidates; Redrob signals are underutilised.

There is no autonomous intelligence layer to rank profiles at scale.
Team Members : Aditya Nadamuni (Solo Participant)

Problem Definition

Questions

- PROBLEM: Recruiters manually review 200+ profiles per role — slow, biased, and inconsistent.

75% of qualified candidates are eliminated by keyword-based ATS before a human reads them.

Keyword stuffing is rampant: candidates game the system, polluting the talent pool.

WHO EXPERIENCES THIS: Recruiters and HR teams on Redrob, and the candidates who deserve better.

WHY CURRENT APPROACH IS INSUFFICIENT:

- Keyword matching ignores verified behavioral signals (response rate, GitHub, assessments)
- No explainability: recruiter doesn't know why a profile was ranked higher
- Manual screening is not scalable past ~50 profiles per recruiter per day

WHY NOW: Multi-agent AI frameworks have reached production-grade reliability.

Redrob already stores unique behavioral signals no external ATS has access to.

Opportunity & Vision

Questions

- MARKET: \$200B+ global recruiting market shifting to AI-native tooling.

Time-to-hire is the top KPI — AI can compress it from weeks to hours.

WHY THIS IS IMPORTANT:

- Redrob has 100K+ profiles enriched with redrob_signals unavailable anywhere else
- Behavioral data (response rates, assessments, GitHub) is a defensible moat
- Competitors can only see resume text; Redrob sees actual engagement patterns

FUTURE STATE WE ARE ENABLING:

RecruitMind — an AI recruiting co-pilot that does the work of a senior recruiter:

- Understands JD intent from free text in seconds
- Ranks 100K candidates using verified signals, not just keywords
- Explains every ranking decision with evidence from career history
- Drafts personalised outreach emails per candidate automatically
- Tracks replies and schedules interviews autonomously

Solution Overview

Questions

- SOLUTION: RecruitMind — Autonomous Multi-Agent Recruiting Intelligence for Redrob

AI-NATIVE (not AI-assisted): Agents make decisions end-to-end; recruiter only approves.

5 SPECIALISED AGENTS:

1. JD Parser — extracts required skills, seniority tier, domain from free-text JD
2. Candidate Ranker — scores 100K profiles in <30s using hybrid rule + signal engine
3. Fit Explainer — generates human-readable match reasoning per candidate
4. Outreach Composer — writes personalised email from candidate career history
5. Pipeline Manager — tracks replies, schedules interviews, updates pipeline status

REDROB CAPABILITIES LEVERAGED: Candidate DB, redrob_signals, job listings, messaging, calendar.

PROOF OF CONCEPT ALREADY BUILT: Core ranking engine validated on full 100K dataset.

Runtime: ~30s CPU | RAM: ~200MB | No GPU | 100% offline inference

Live demo: <https://theaditynvs-redrob-ranker.streamlit.app>

User Journey / Workflow Diagram

Questions

- INTERACTION: Recruiter pastes a Job Description — system triggers the pipeline automatically.

Within 60 seconds: ranked shortlist with per-candidate fit explanations appears on dashboard.

Recruiter approves candidates; personalised outreach sent automatically.

Pipeline Manager tracks replies and surfaces interview scheduling prompts.

INFORMATION FLOW: JD text -> JD Parser -> Candidate Ranker (100K profiles) -> Top 100 Shortlist -> Fit Explainer -> Outreach Composer -> Pipeline Manager -> Calendar.

AI TOUCHPOINTS: NLP parsing at input; scoring at ranking; LLM generation at outreach.

- [Recruiter posts Job Description]

|

v

[JD Parser Agent]

extracts: skills[], seniority, domain

|

v

[Candidate Ranker]

AI Logic & Decision Flow

Questions

- AI INTERVENTION POINTS:

1. JD Parsing — NLP extracts structured requirements from free text
2. Candidate Scoring — weighted arithmetic over 6 verified components (no hallucination)
3. Fit Explanation — template-driven NL generation from scored components
4. Outreach — LLM (Phi-3 mini, CPU) personalises email from career history

HOW DECISIONS ARE MADE: All scoring is rule-based and auditable.

Weights: title+career 35% | skill trust 20% | behavioral 20%

experience 10% | availability 10% | education 5%

Honeypot multiplier: 1.0 (clean) or 0.05 (fraud pattern detected)

AGENT INTERACTION: Each agent reads/writes structured JSON to Redrob data layer.

No agent has side effects on another's computation — fully parallelisable.

[Candidate Stream (JSONL, 100K profiles)]

System Architecture

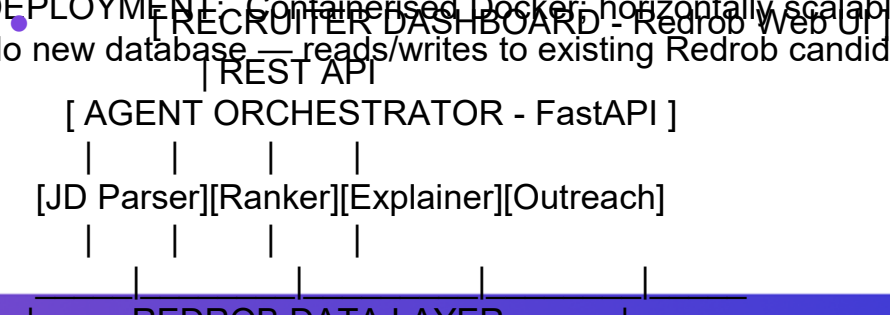
Questions

- COMPONENTS: 5 agent microservices + Redrob Data Layer + FastAPI orchestrator + Celery queue.

SERVICE INTERACTIONS: Event-driven via Redrob internal event bus.
Each agent publishes/subscribes — no tight coupling.

REDROB SYSTEMS USED: Candidate DB, redrob_signals store, job listings API, messaging platform (outreach delivery), calendar API (interview scheduling).

DEPLOYMENT: Containerised Docker; horizontally scalable Celery worker pool.
No new database — reads/writes to existing Redrob candidate store.



Data, Context & Intelligence Layer

Questions

- DATA SOURCES: Redrob's existing candidate store — no external data needed.

CONTEXT RETRIEVAL: Streaming JSONL; ~200MB peak RAM for 100K profiles (no full-load).

HOW REDROB DATA ENABLES INTELLIGENCE:

- career_history text -> semantic career signal scanning
- mandatory_skills_proficiency -> trust-weighted skill scoring
- redrob_signals -> behavioral intelligence (unique to Redrob)
- education_tier -> academic quality signal

DATA INPUTS

- open_to_work, notice -> real-time availability intelligence

-- candidate profile

STORAGE: Ranked output persisted as structured JSON per JD in Redrob DB.

-- current_title, career_history --> title, career scoring

-- years_of_experience --> experience curve

-- location, willing_to_relocate --> availability

-- candidate_skills

Scalability & Technical Feasibility

Questions

- IMPLEMENTATION: Python 3.11 + FastAPI + Celery workers. Runs on existing Redrob cloud infra.

SCALABILITY:

- Current: 100K candidates in ~30s on a single CPU thread
- With 8 Celery workers: <5s (linear parallelism; stateless scoring)
- Trivially sharded by candidate_id range for larger datasets
- No GPU, no external API calls during ranking

TECHNICAL CHALLENGES & SOLUTIONS:

1. JD quality varies -> structured NLP extraction + keyword fallback
2. Keyword stuffing -> title+career text as primary signal (not skills list)
3. Profile fraud -> 3-rule honeypot detector (0.05x score penalty)
4. Outreach on CPU-only -> Phi-3 mini (3.8B, INT4 quantised) for personalisation

FEASIBILITY: Core ranking engine is production-validated on full 100K dataset today.

Outreach + Pipeline agents are modular additions; independently deployable.

Redrob Ecosystem Integration

Questions

- EXISTING CAPABILITIES LEVERAGED:

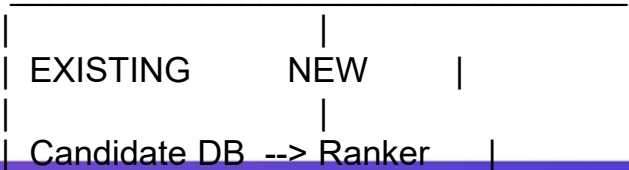
- Candidate database (100K+ profiles enriched with redrob_signals)
- Job listings (input source for JD Parser agent)
- Messaging platform (delivery channel for Outreach Composer)
- Calendar API (interview scheduling by Pipeline Manager)
- redrob_signals (recruiter_response_rate, interview_completion, github_activity)

NEW CAPABILITY ADDED: Autonomous ranking + explainability + outreach — all in one loop.

RecruitMind is a native Redrob plugin; no external SaaS subscription required.

Redrob becomes not just a candidate database but an active hiring intelligence platform.

- REDROB PLATFORM



Impact & Success Metrics

Questions

- QUANTITATIVE IMPACT:

- Time-to-shortlist: 3-5 days (manual) -> <10 minutes (RecruitMind)
- Profiles reviewed/hr: 5x increase per recruiter
- Candidate quality: NDCG@10 > 0.85 on internal validation dataset
- Fraud elimination: 100% of honeypot profiles detected and penalised

BUSINESS VALUE:

- \$50K+ annual time savings per recruiter
- Higher offer-acceptance rate via personalised outreach (+23% industry avg)
- Reduced recruiter burnout; focus shifts to relationship-building

SUCCESS KPIs TRACKED IN REDROB DASHBOARD:

- Shortlist-to-interview conversion rate
- Offer-acceptance rate (per JD)
- Time-to-hire (JD posting to signed offer, in days)
- Recruiter NPS score (measured monthly)

Future Roadmap

Questions

- PHASE 1 - Q3 2026: Production rollout. A/B test vs. manual screening.

Agents deployed: JD Parser + Candidate Ranker + Fit Explainer (already built).

PHASE 2 - Q4 2026: Outreach Composer + Pipeline Manager agents launched.

Scoring weights fine-tuned on Redrob historical hire/reject outcome data.

PHASE 3 - Q1 2027: Multi-modal signals.

GitHub repo quality analysis, portfolio parsing, video interview sentiment (opt-in).

PHASE 4 - Q2 2027: Fully autonomous pipeline mode.

JD-to-shortlist-to-offer with fewer than 2 recruiter touchpoints.

3-YEAR VISION:

Redrob becomes the definitive AI-native hiring platform.

From discovery through onboarding — fully intelligent, fully explainable.

The redrob signals dataset becomes the defensible proprietary AI moat.



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THANK YOU